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Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Code: 100308 ***Time: 03 Hours***

Semester: III ***Subject: Electromagnetic Fields*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) The dot product of two vectors is zero when:
- (i) They are parallel
- (ii) One is a scalar
- (iii) They are perpendicular
- (iv) Their magnitudes are equal

(b) Coulomb's law describes the force between:
- (i) Two magnetic poles
- (ii) Two stationary point charges
- (iii) Two current-carrying wires
- (iv) A moving charge and a magnetic field

(c) Current density **J** is related to current *I* by:
- (i) $I = \mathbf{J} \cdot \mathbf{A}$
- (ii) $\mathbf{J} = I^2\mathbf{A}$
- (iii) $I = \mathbf{E} \cdot \mathbf{A}$
- (iv) $\mathbf{J} = I / A^2$

(d) The unit of magnetic flux is:
- (i) Tesla
- (ii) Farad
- (iii) Henry
- (iv) Weber

(e) The force on a moving charge in a magnetic field is given by:
- (i) $\mathbf{F} = q\mathbf{E}$
- (ii) $\mathbf{F} = q\mathbf{B}$
- (iii) $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$
- (iv) $\mathbf{F} = \mathbf{v} \times \mathbf{E}$

(f) Faraday's law of electromagnetic induction states that:
- (i) Magnetic field induces electric current
- (ii) Changing electric field induces magnetic field
- (iii) Changing magnetic flux induces electromotive force
- (iv) Magnetic field is proportional to current

(g) The wave equation in free space is derived from:
- (i) Ohm's law
- (ii) Faraday's law only
- (iii) Biot–Savart Law
- (iv) Maxwell's equations

(h) The cross product of two vectors is a:
- (i) Vector
- (ii) Scalar
- (iii) Tensor
- (iv) Matrix

(i) The relative permittivity ε_r of a material affects:
- (i) Current in a conductor
- (ii) Capacitance of a capacitor
- (iii) Magnetic field strength
- (iv) Resistance of a wire

(j) The displacement current was introduced by Maxwell to:
- (i) Account for magnetic field in conductors
- (ii) Explain magnetic flux in insulators
- (iii) Ensure continuity of current in a capacitor
- (iv) Explain dielectric heating

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Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Describe the procedure for converting a vector from one coordinate system from cylindrical to rectangular coordinates. Show all necessary steps with an example.

(b) State and explain Gauss's Divergence Theorem and Stokes' Theorem.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Define electric field intensity. Derive an expression for the electric field due to a point charge?

(b) Derive the expression for the potential difference in the following configurations:
(i) Between two point charges
(ii) Due to a uniformly charged ring on its axis

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Define capacitance. Derive an expression for the capacitance of a parallel plate capacitor with dielectric.

(b) State and derive Laplace's and Poisson's equations from Gauss's law. Under what conditions does each apply?

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Define magnetic flux and magnetic flux density. Derive the relation between **B** and **H**?

(b) Derive the expression for the magnetic field produced by a finite length straight conductor using Biot-Savart Law.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Derive the force experienced by a differential current element placed in a magnetic field.

(b) Describe the different types of magnetic materials (diamagnetic, paramagnetic, and ferromagnetic) and their behavior in an external magnetic field.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Write all four Maxwell's equations in both point form and integral form. Briefly describe the physical meaning of each.

(b) Derive the expression for motional EMF when a conductor moves in a magnetic field.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) State and prove the Poynting theorem.

(b) What is the skin effect? Derive the expression for skin depth and explain its dependence on frequency and material properties.

Q.9 Solve both questions : **[7 x 2 = 14]**

(a) Define current and current density. Derive the relationship between them and explain how current density varies in a non-uniform conductor.

(b) Define magnetization and magnetic susceptibility. Derive the relation between **B**, **H** and **M**.

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